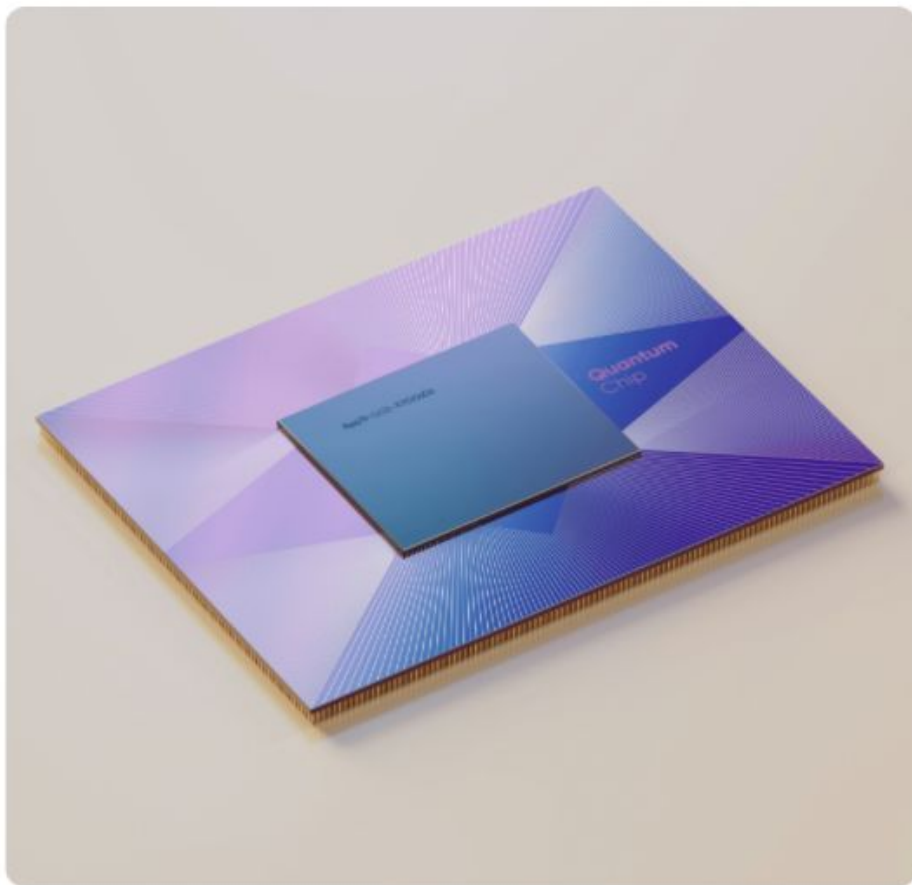


Low-Cost IoT Based Motor Efficiency and Safety Monitoring System for Household Appliances

Low-Cost IoT: Motor Efficiency and Safety



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Project Title and Objectives

Project Title

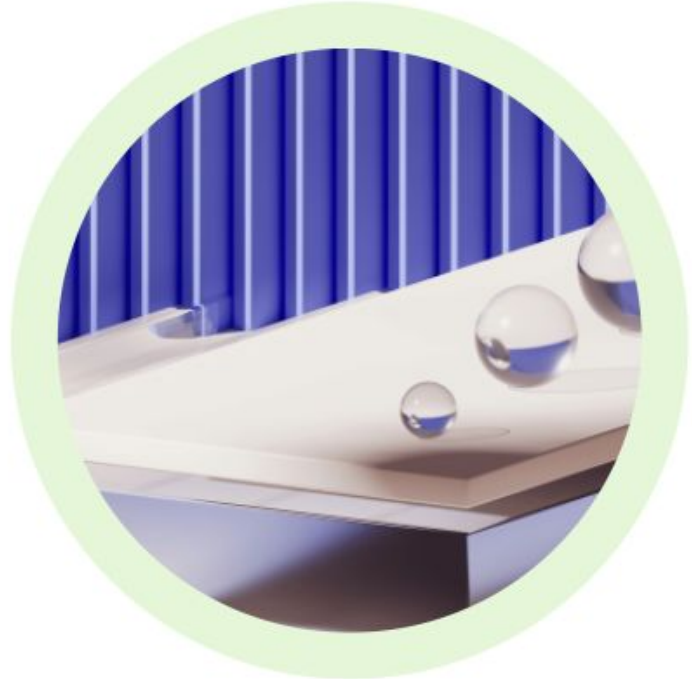
Low-Cost IoT Based Motor Efficiency and Safety Monitoring System for Household Appliances

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Primary Objective

Design and implement a low-cost IoT-based monitoring system that continuously evaluates the health, efficiency, and safety of household electric motors such as water pumps, fans, and washing machines. The system measures electrical and thermal parameters, calculates power consumption and efficiency degradation, and transmits data to a cloud platform for real-time monitoring and early fault detection.



System Components Required

Central Processing

ESP32 Microcontroller: Central processing unit for data acquisition, processing, and Wi-Fi communication

Sensing & Monitoring

ACS712 Current Sensor: Measures motor current consumption in real-time for load analysis and efficiency calculation

Voltage Sensor: Monitors supply voltage variations and detects voltage anomalies affecting motor performance
Temperature Sensor: Continuously tracks motor thermal conditions to detect overheating and degradation

Safety & Control

Relay Module: Provides safety protection by enabling automatic motor shutdown upon detection of abnormal conditions

Power Management

Power Supply Unit: Provides stable power to the monitoring system and peripheral components

Target Device

Household Motor: Target device for monitoring (water pumps, fans, washing machines, etc.)

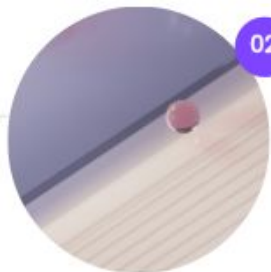
System Block Diagram and Architecture



01

Sensor Layer

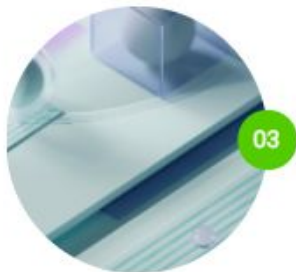
ACS712 current sensor, voltage sensor, and temperature sensor collect real-time operational parameters from the household motor.



02

Processing Layer

ESP32 microcontroller receives analog signals from sensors, performs analog-to-digital conversion, processes data using signal conditioning and mathematical algorithms to calculate power consumption, efficiency metrics, and detect anomalies.



03

Communication and Cloud Layer

Collected data is transmitted via Wi-Fi protocol to a cloud-based IoT platform where data is stored, visualized, and accessed through mobile or web dashboards for remote monitoring. The relay module is controlled by the ESP32 to safely disconnect the motor during fault conditions.

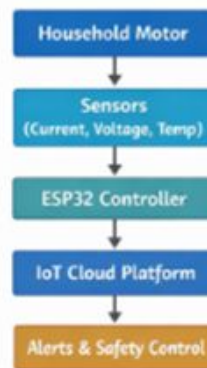


04

Data Flow

Sensors → ESP32 ADC → Signal Processing → Anomaly Detection → Wi-Fi Transmission → Cloud Storage → User Dashboard & Alerts

IoT Motor Efficiency Monitoring System



Key Functionalities and Features

01

Real-Time Parameter Monitoring

The system continuously measures voltage, current, and temperature with appropriate sensor resolution and sampling rates to capture motor operational dynamics accurately.

02

Power Consumption Calculation

Calculates instantaneous and average power consumption using voltage and current measurements: $P = V \times I$. Tracks cumulative energy usage over time periods.

03

Automatic Safety

Upon detection of critical faults, the system activates a relay module to safely disconnect the motor from power supply, preventing equipment damage and potential safety hazards.

04

Efficiency Degradation Detection

Compares current efficiency metrics against baseline values to identify early signs of motor aging, bearing wear, or winding degradation. Efficiency is estimated as: $\eta = (\text{Output Power}) / (\text{Input Power})$, with degradation indicated by progressive efficiency loss.

05

Cloud-Based Monitoring

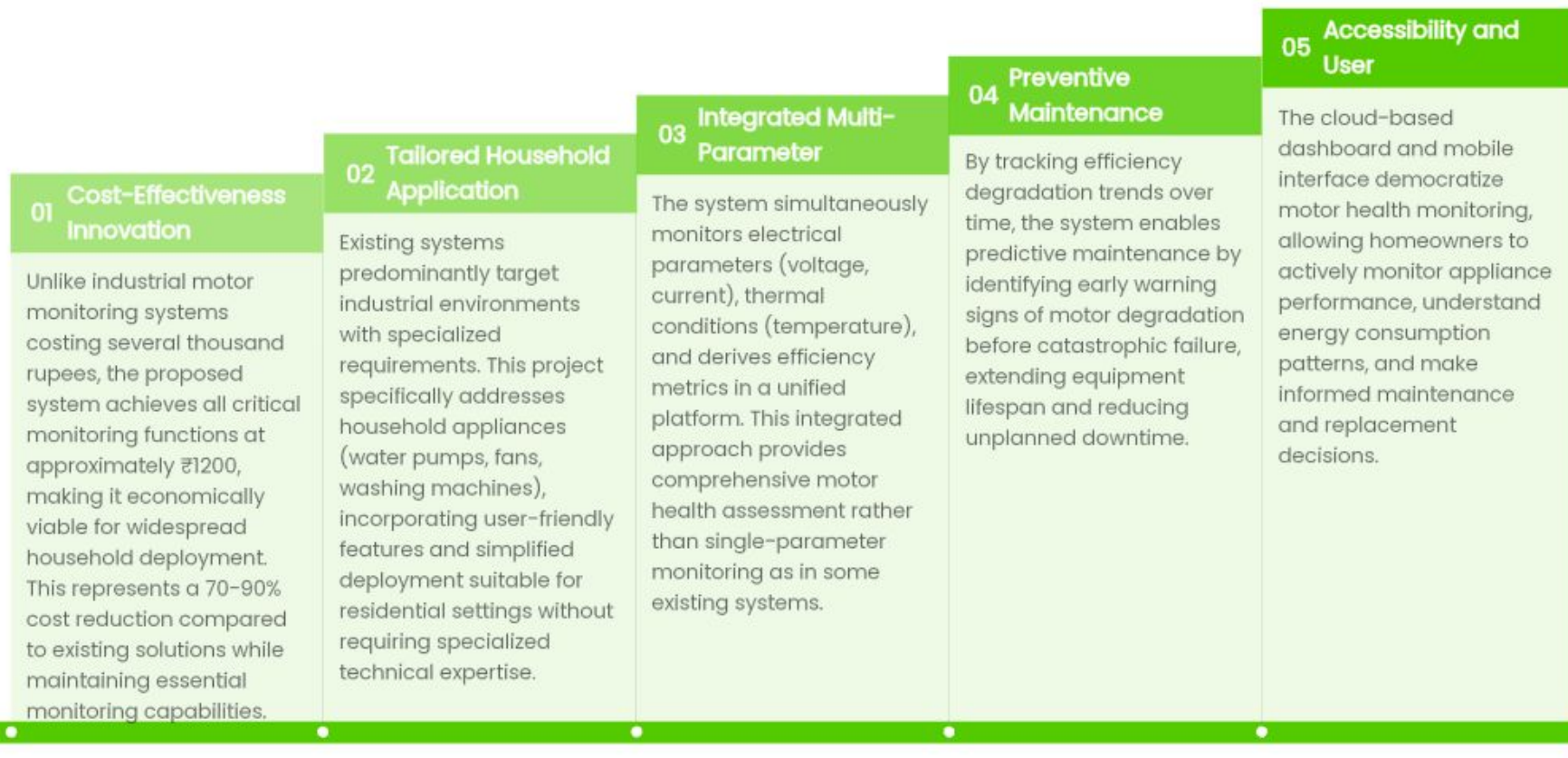
Transmits processed data through Wi-Fi to cloud IoT platform for real-time visualization, historical data analysis, trend monitoring, and predictive maintenance insights accessible via mobile and web interfaces.

06

Anomaly Detection

Monitors for abnormal conditions including motor overload (excessive current draw), overheating (temperature exceeding safe thresholds), and efficiency drop (indicating mechanical or electrical faults). Generates real-time alerts via cloud platform notification system.

Novelty and Project Innovation



Literature Review Table

Review Summary

Literature Review Summary: The following table presents key research works on motor health monitoring systems, their technical approaches, and relevance to the proposed project:

Key Research Works

The table below documents the evolution of motor monitoring technology from industrial-scale systems to the proposed low-cost household appliance solution, highlighting the technical contributions adapted from existing research.

Reference	Research Paper Title	Technical Approach	Key Parameters Monitored	Communication Method	Application Domain	Contributions Adopted in Proposed
1	Wireless Health Monitoring	Three-axis accelerometer for	Vibration signals, mechanical	ZigBee wireless sensor	Industrial motor monitoring	Wireless data transmission
2	IoT Based Health Monitoring	Multi-sensor integration (temperature,	Temperature, vibration, electrical	Wi-Fi to cloud platform	Industrial and general motors	Wi-Fi communication protocol;
3	Digitalized Condition Monitoring	Digital signal processing.	Operational parameters (multiple).	Data acquisition and	Industrial motor health management	Signal processing techniques;
4	Essential Motor Health Monitoring	Condition-based monitoring	Temperature, vibration, electrical	Standard operating limit	General motor monitoring	Real-time parameter threshold
5	Intelligent Condition Monitoring	Programmable Logic Controller	Motor operational conditions,	PLC-based local control system	Industrial environments	Signal conditioning techniques;
6	Low-Cost IoT Based Motor	ESP32 micro controller, multiple	Voltage, current, temperature,	Wi-Fi to cloud IoT platform	Household motor applications	Adapted for household scale: Wi-Fi

Current Working Status and Progress

Comparison Item A

Phase 1 – Design and Planning

System architecture finalized; component selection completed based on cost-performance analysis; circuit schematics designed for sensor integration with ESP32 microcontroller; Wi-Fi IoT cloud platform selected and evaluated; relay-based motor protection circuit designed.

Phase 2 – Hardware Assembly

ESP32 development board configured; ACS712 current sensor calibrated and integrated into power monitoring circuit; voltage sensor circuit constructed and tested; temperature sensor (DS18B20 or LM35) integrated with ADC input; relay module connected to GPIO output for motor control; signal conditioning circuits implemented for analog sensor outputs.

Phase 3 – Firmware Development

ESP32 firmware developed in Arduino IDE; sensor reading routines implemented with appropriate sampling rates; analog-to-digital conversion code configured; power consumption calculation algorithms programmed; efficiency degradation detection logic implemented; Wi-Fi connectivity module integrated; cloud API communication established; alert threshold logic implemented; relay control logic incorporated.



Comparison Item B

Phase 4 – Cloud Platform

IoT cloud platform (ThingSpeak, Blynk, or AWS IoT) configured; data storage schema designed; real-time data visualization dashboard created; mobile app integration initiated; alert notification system configured; historical data analysis features implemented.

Phase 5 – Testing

Laboratory testing with household motor test bench; sensor accuracy validation against calibrated instruments; efficiency calculation verification; alert system functionality testing; relay protection mechanism testing; field trials with actual household appliances; system reliability and durability assessment.

Completion Timeline

Full system deployment and household testing targeted for completion within the academic semester; final documentation and project demonstration scheduled.



 Company Name

Thanks



For your kind attention.